



DATABASES PROJECT FINAL REPORT

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DATABASE INITIAL STUDY AND ANALYSIS

DATA GATHERING METHODS

In order to identify and understand the information and data requirements of the R&D Directorate at Kadir Has University, the team applied a combination of conceptual and document-based data gathering methods. Since direct access to real stakeholders was not feasible, the following approaches were used:

Use Case Analysis:

The primary source of requirements was the detailed use case scenario provided in the project description. The team carefully analyzed the operational workflow of the R&D Directorate, including project registration and assignment, funding management, expense tracking, and deliverable monitoring.

Procedure and Document Analysis:

The team examined typical university research administration procedures, such as grant allocation, project coordination, academic staff involvement, and financial reporting. This helped in understanding how similar directorates operate and what types of data are commonly recorded.

Analytical Reasoning and Assumptions:

Based on the collected information, reasonable assumptions were made to simplify real-world complexities. These assumptions ensured that the database design remains manageable while still supporting essential business operations and decision-making needs.

INFORMATION REQUIREMENTS OF THE SYSTEM

Organizational Overview

The R&D Directorate at Kadir Has University is responsible for managing and monitoring all research projects conducted within the university. Its main objectives include:

- Registering and tracking research projects from initiation to completion
- Monitoring funding and financial transactions related to projects
- Managing academic staff involvement and workload
- Tracking project outputs and deliverables
- Supporting reporting and decision-making for management and funding bodies

Actors / Roles

The primary actors of the system include:

- R&D Directorate Managers:
 - Oversee projects, funding, expenses, and reporting obligations.
- Project Coordinators (Academic Staff):
 - Responsible for leading projects, assigning roles, and ensuring project progress.
- University Management:
 - Analyze research activity and funding distribution.
- Financial Officers:
 - Monitor incoming funds, expenses, and budget utilization.

Business Operations

Based on the use case, the system supports the following key operations:

- Registering research projects with details such as title, status, duration, budget, and coordinator
- Assigning academic staff to projects with defined roles and effort levels
- Recording funding received from different funding bodies
- Tracking project-related expenses
- Monitoring deliverables and their submission and approval status
- Classifying projects by scientific fields for analytical purposes

User Information Requirements

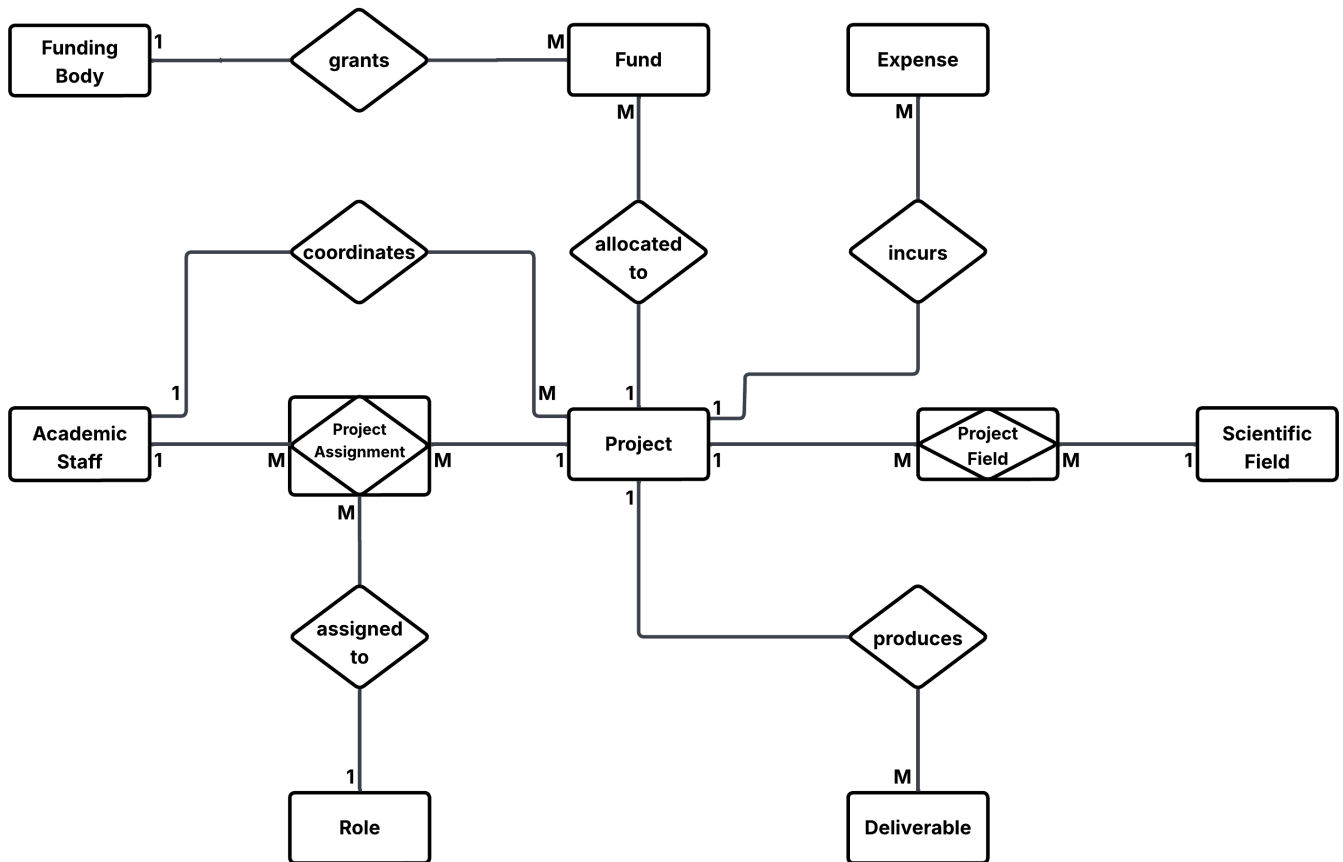
To support decision-making and reporting, the system must be able to provide the following information through queries and reports:

- A list of active projects and their project coordinators
- The number of academic staff members working on each project
- The workload of academic staff across multiple projects
- Total working hours contributed by staff members to specific projects
- Total funding received per project and per scientific field
- Total expenses per project and comparison with the approved budget
- Remaining budget for each project
- Deliverables produced by projects and their approval status
- Distribution of projects across scientific fields
- Identification of projects funded by multiple funding bodies

Scope and Simplifications: Due to the complexity of real-world research management systems, the database design focuses on core operations only. However, the proposed design sufficiently supports the most critical information needs of the R&D Directorate.

DATABASE DESIGN: CONCEPTUAL DESIGN

ENTITY-RELATIONSHIP DIAGRAM



This figure shows the Entity–Relationship (ER) model of the system, developed using Chen notation.

BUSINESS RULES

- **Funding Body – Fund**
Each Funding Body grants **one or many** Funds.
Each Fund is granted by **one** Funding Body.
- **Fund – Project**
Each Project can receive **one or many** Funds.
Each Fund can be allocated to **one** Project.
- **Expense – Project**
Each Project incurs **one or many** Expenses.
Each Expense is incurred by **one** Project.
- **Project Field (Scientific Field – Project)**
Each Project is associated with **one or many** Scientific Fields.
Each Scientific Field is associated with **one or many** Projects.

- **Deliverable – Project**
Each Project produces **one or many** Deliverables.
Each Deliverable is produced by **one** Project.
- **Academic Staff – Project**
Each Project is coordinated by **one** Academic Staff member (Project Coordinator).
Each Project Coordinator can coordinate **one or many** Projects.
- **Project Assignment (Academic Staff – Project)**
Each Project is worked on by **one or many** Academic Staff members.
Each Academic Staff member can work on **one or many** Projects
- **Project Assignment – Role**
Each Project Assignment is assigned **one** Role.
Each Role can be associated with **many** Project Assignments.

DATA DICTIONARY

Table Name	Attribute Name	Contents	Type	Format	Domain	Required	PK / FK	FK Referenced Table
funding_body	funding_body_id	Funding body identifier	CHAR(4)	XX99	FB01-FB99	Y	PK	
	name	Funding body name	VARCHAR(30)	Xxxxxxxxxx		Y		
	country	Country of funding body	VARCHAR(30)	Xxxxxxxxxx		N		
	type	Type of funding body	VARCHAR(20)	Xxxxxxxxxx		N		
fund	fund_id	Fund identifier	CHAR(6)	X99999	F00001-F99999	Y	PK	
	amount	Fund amount	NUMERIC(12,2)	9,999,999.99	≥ 0	Y		
	received_date	Date fund received	DATE	dd-mm-yyyy	Valid date	Y		
	funding_body_id	Granting funding body	CHAR(4)	XX99	FB01-FB99	Y	FK	funding_body
	project_id	Project allocated to	CHAR(5)	X9999	P0001-P9999	Y	FK	project
project	project_id	Project identifier	CHAR(5)	X9999	P0001-P9999	Y	PK	

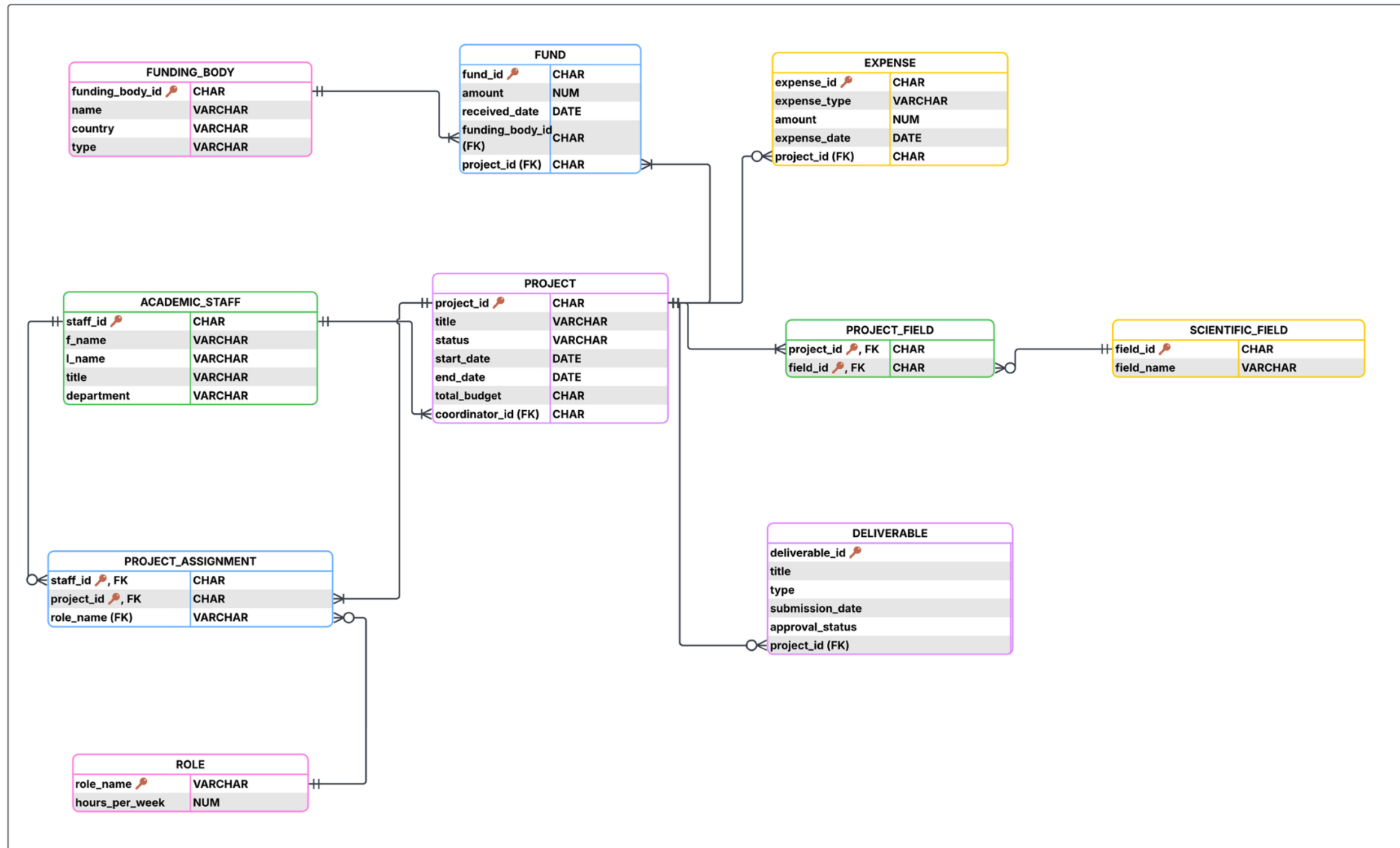
Table Name	Attribute Name	Contents	Type	Format	Domain	Required	PK / FK	FK Referenced Table
	title	Project title	VARCHAR(50)	Xxxxxxxxxx		Y		
	status	Project status	VARCHAR(30)	Xxxxxxxxxx	Active, Completed, Suspended	Y		
	start_date	Project start date	DATE	dd-mm-yyyy	Valid date	Y		
	end_date	Project end date	DATE	dd-mm-yyyy	Valid date	N		
	total_budget	Approved budget	NUMERIC(12,2)	9,999,999.99	≥ 0	Y		
	coordinator_id	Project coordinator	CHAR(4)	X999	A001–A999	Y	FK	academic_staff
expense	expense_id	Expense identifier	CHAR(6)	X99999	E00001-E99999	Y	PK	
	expense_type	Type of expense	VARCHAR(30)	Xxxxxxxxxx		Y		
	amount	Expense amount	NUMERIC(10,2)	9,999,999.99	≥ 0	Y		
	expense_date	Expense date	DATE	dd-mm-yyyy	Valid date	Y		
	project_id	Project related	CHAR(5)	X9999	P0001–P9999	Y	FK	project
academic_staff	staff_id	Staff identifier	CHAR(4)	X999	A001–A999	Y	PK	

Table Name	Attribute Name	Contents	Type	Format	Domain	Required	PK / FK	FK Referenced Table
	f_name	First name	VARCHAR(30)	Xxxxxxxxxx		Y		
	l_name	Last name	VARCHAR(30)	Xxxxxxxxxx		Y		
	title	Academic title	VARCHAR(30)	Xxxxxxxxxx		Y		
	department	Department name	VARCHAR(30)	Xxxxxxxxxx		N		
role	role_name	Role identifier	VARCHAR(30)	Xxxxxxxxxx		Y	PK	
	hours_per_week	Hours worked	NUMERIC(2)	99	0 – 40	Y		
project_assignment	staff_id	Assigned staff member	CHAR(4)	X999	A001–A999	Y	PK, FK	academic_staff
	project_id	Assigned team	CHAR(5)	X9999	P0001–P9999	Y	PK, FK	project
	role	Assigned role	VARCHAR(30)	Xxxxxxxxxx		Y	FK	role
deliverable	deliverable_id	Deliverable identifier	CHAR(6)	X99999	D00001–D99999	Y	PK	
	title	Deliverable title	VARCHAR(100)	Xxxxxxxxxx		Y		

Table Name	Attribute Name	Contents	Type	Format	Domain	Required	PK / FK	FK Referenced Table
	type	Deliverable type	VARCHAR(30)	Xxxxxxxxxx		Y		
	submission_date	Submission date	DATE	dd-mm-yyyy	Valid date	Y		
	approval_status	Approval status	VARCHAR(20)	Xxxxxxxxxx	Submitted, Approved, Rejected	Y		
	project_id	Associated project	CHAR(5)	X9999	P0001–P9999	Y	FK	project
scientific_field	field_id	Field identifier	CHAR(3)	X99	S01-S99	Y	PK	
	field_name	Scientific field name	VARCHAR(30)	Xxxxxxxxxx		Y		
project_field	project_id	Related project	CHAR(5)	X9999	P0001–P9999	Y	PK, FK	project
	field_id	Related field	CHAR(3)	X99	S01-S99	Y	PK, FK	scientific_field

DATABASE DESIGN: LOGICAL DESIGN

RELATIONAL SCHEMA



PHYSICAL IMPLEMENTATION AND QUERYING

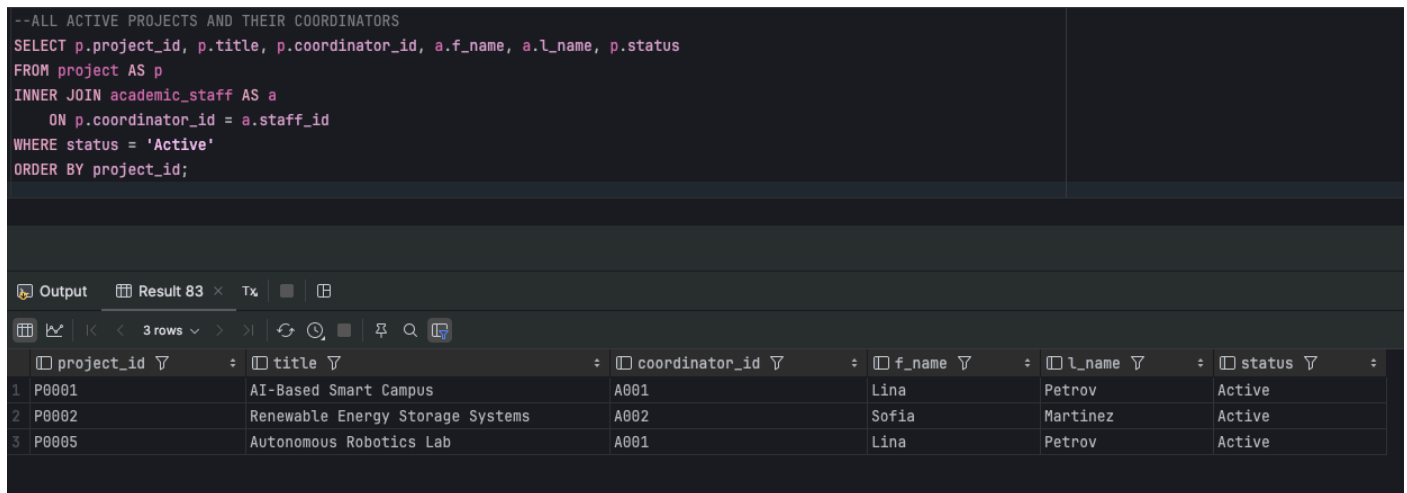
The database was implemented using PostgreSQL running in a Docker container. It is submitted as a PostgreSQL SQL dump (rnd_directorate.sql).

QUERY 1: ALL ACTIVE PROJECTS AND THEIR COORDINATORS

Query:

```
SELECT p.project_id, p.title, p.coordinator_id, a.f_name, a.l_name, p.status
FROM project AS p
INNER JOIN academic_staff AS a
    ON p.coordinator_id = a.staff_id
WHERE status = 'Active'
ORDER BY project_id;
```

Output:



The screenshot shows a PostgreSQL query execution interface. The query is repeated at the top: `--ALL ACTIVE PROJECTS AND THEIR COORDINATORS` followed by the same SELECT statement as above. Below the query, the results are displayed in a table with 6 columns: project_id, title, coordinator_id, f_name, l_name, and status. There are 3 rows of data.

project_id	title	coordinator_id	f_name	l_name	status
P0001	AI-Based Smart Campus	A001	Lina	Petrov	Active
P0002	Renewable Energy Storage Systems	A002	Sofia	Martinez	Active
P0005	Autonomous Robotics Lab	A001	Lina	Petrov	Active

Query Specifications:

Who will use the query:

R&D Directorate Managers

Purpose of the query:

To identify all currently active research projects and the academic staff members coordinating them.

What the query displays:

This query displays the project ID, project title, coordinator ID, coordinator's first and last name, and project status for all projects that are marked as Active. The results are ordered by project ID.

QUERY 2: NUMBER OF TEAM MEMBERS PER PROJECT WHERE THERE IS MORE THAN ONE MEMBER

Query:

```
SELECT p.project_id, p.title, COUNT(pa.staff_id) AS num_of_members
FROM project AS p
JOIN project_assignment AS pa
    ON p.project_id = pa.project_id
```

```
GROUP BY p.project_id, p.title
HAVING COUNT(pa.staff_id) > 1
ORDER BY p.project_id;
```

Output:

```
--NUMBER OF TEAM MEMBERS PER PROJECT WHERE THERE IS MORE THAN ONE MEMBER
SELECT p.project_id, p.title, COUNT(pa.staff_id) AS num_of_members
FROM project AS p
JOIN project_assignment AS pa
  ON p.project_id = pa.project_id
GROUP BY p.project_id, p.title
HAVING COUNT(pa.staff_id) > 1
ORDER BY p.project_id;
```

Output Result 84 Tx

5 rows

	project_id	title	num_of_members
1	P0001	AI-Based Smart Campus	6
2	P0002	Renewable Energy Storage Systems	4
3	P0003	Genomic Data Analysis Platform	3
4	P0004	Hydrogen Energy Research	3
5	P0005	Autonomous Robotics Lab	3

Query Specifications:

Who will use the query:

R&D Directorate Managers

Purpose of the query:

To analyze project team sizes and identify projects involving more than one academic staff member.

What the query displays:

This query shows the project ID, project title, and the total number of assigned academic staff members for each project that has more than one team member.

QUERY 3: STAFF MEMBERS AND NUMBER OF PROJECTS THEY ARE ASSIGNED TO

Query:

```
SELECT a.staff_id, a.f_name, a.l_name, a.title, a.department, COUNT(pa.staff_id) AS
num_projects
FROM academic_staff AS a
JOIN project_assignment AS pa
  ON a.staff_id = pa.staff_id
GROUP BY a.staff_id, a.title, a.department, a.f_name, a.l_name
ORDER BY a.staff_id;
```

Output:

```
--STAFF MEMBERS AND NUMBER OF PROJECTS THEY ARE ASSIGNED TO
SELECT a.staff_id, a.f_name, a.l_name, a.title, a.department, COUNT(pa.staff_id) AS num_projects
FROM academic_staff AS a
JOIN project_assignment AS pa
    ON a.staff_id = pa.staff_id
GROUP BY a.staff_id, a.title, a.department, a.f_name, a.l_name
ORDER BY a.staff_id;
```

	staff_id	f_name	l_name	title	department	num_projects
1	A001	Lina	Petrov	Professor	Computer Engineering	2
2	A002	Sofia	Martinez	Professor	Electrical Engineering	1
3	A003	Elena	Rossi	Professor	Bioengineering	1
4	A004	Min-Jae	Park	Postdoctoral Fellow	Computer Engineering	1
5	A005	Priya	Sharma	PhD Student	Industrial Engineering	2
6	A006	Samuel	Okoye	Professor	Energy Systems	1
7	A007	Maria	Gonzalez	Assistant Professor	Computer Engineering	2
8	A008	Omar	Hassan	Postdoctoral Fellow	Electrical Engineering	2
9	A009	Alina	Smith	Assistant Professor	Industrial Engineering	1
10	A011	Fatma	Celik	PhD Student	Bioengineering	1
11	A012	Li	Wei	Postdoctoral Fellow	Data Science	2
12	A013	Emre	Sahin	Research Assistant	Computer Engineering	1
13	A014	Anna	Novak	PhD Student	Energy Systems	2

Query Specifications:

Who will use the query:

R&D Directorate Managers and Project Coordinators

Purpose of the query:

To assess academic staff involvement across multiple projects.

What the query displays:

This query lists each academic staff member along with their personal details and the total number of projects they are assigned to.

QUERY 4: STAFF MEMBERS AND TOTAL HOURS WORKED ACROSS ALL PROJECTS TILL CURRENT DATE

Query:

```
SELECT a.staff_id, a.f_name, a.l_name, a.title, a.department,
       ((SELECT SUM(r.hours_per_week * ((COALESCE(p.end_date, CURRENT_DATE) -
p.start_date)/7))
FROM project_assignment pa2
JOIN role r
    ON pa2.role_name = r.role_name
JOIN project p
    ON pa2.project_id = p.project_id
WHERE pa2.staff_id = a.staff_id) AS total_hours_worked
FROM academic_staff a
LEFT JOIN project_assignment pa
```

```

    ON a.staff_id = pa.staff_id
GROUP BY a.staff_id, a.f_name, a.l_name, a.title, a.department
ORDER BY a.staff_id;

```

Output:

```

--STAFF MEMBERS AND TOTAL HOURS WORKED ACROSS ALL PROJECTS TILL CURRENT DATE
SELECT a.staff_id, a.f_name, a.l_name, a.title, a.department,
       (SELECT SUM(r.hours_per_week * ((COALESCE(p.end_date, CURRENT_DATE) - p.start_date)/7))
        FROM project_assignment pa2
         JOIN role r
           ON pa2.role_name = r.role_name
         JOIN project p
           ON pa2.project_id = p.project_id
        WHERE pa2.staff_id = a.staff_id) AS total_hours_worked
FROM academic_staff a
LEFT JOIN project_assignment pa
  ON a.staff_id = pa.staff_id
GROUP BY a.staff_id, a.f_name, a.l_name, a.title, a.department
ORDER BY a.staff_id;

```

staff_id	f_name	l_name	title	department	total_hours_worked
A001	Lina	Petrov	Professor	Computer Engineering	2660
A002	Sofia	Martinez	Professor	Electrical Engineering	2520
A003	Elena	Rossi	Professor	Bioengineering	2080
A004	Min-Jae	Park	Postdoctoral Fellow	Computer Engineering	1610
A005	Priya	Sharma	PhD Student	Industrial Engineering	3325
A006	Samuel	Okoye	Professor	Energy Systems	3020
A007	Maria	Gonzalez	Assistant Professor	Computer Engineering	3990
A008	Omar	Hassan	Postdoctoral Fellow	Electrical Engineering	8940
A009	Alina	Smith	Assistant Professor	Industrial Engineering	3780
A010	John	Miller	Professor	Robotics	<null>
A011	Fatma	Celik	PhD Student	Bioengineering	2600
A012	Li	Wei	Postdoctoral Fellow	Data Science	5250
A013	Emre	Sahin	Research Assistant	Computer Engineering	920
A014	Anna	Novak	PhD Student	Energy Systems	6925

Query Specifications:

Who will use the query:

R&D Directorate Managers and Financial Officers.

Purpose of the query:

To estimate the total workload of each academic staff member across all projects.

What the query displays:

The query calculates and displays the total number of hours each staff member has worked by multiplying weekly working hours with the project duration (in weeks), considering ongoing projects up to the current date.

QUERY 5: HOW MANY TOTAL HOURS HAS A STAFF MEMBER WORKED FOR A PROJECT

Query:

```

SELECT p.project_id, p.title, pa.staff_id, r.hours_per_week, p.start_date,
       COALESCE(p.end_date, CURRENT_DATE) AS end_date,
       (COALESCE(p.end_date, CURRENT_DATE) - p.start_date) / 7 AS weeks_worked,
       r.hours_per_week * (((COALESCE(p.end_date, CURRENT_DATE) - p.start_date)) / 7) AS
total_hours

```

```
FROM project_assignment AS pa
JOIN role AS r
    ON pa.role_name = r.role_name
JOIN project AS p
    ON pa.project_id = p.project_id
WHERE pa.staff_id = 'A001' AND p.project_id = 'P0001';
```

Output:

```
--HOW MANY TOTAL HOURS HAS STAFF MEMBERS WORKED FOR A PROJECT
SELECT p.project_id, p.title, pa.staff_id, r.hours_per_week, p.start_date,
       COALESCE(p.end_date, CURRENT_DATE) AS end_date,
       (COALESCE(p.end_date, CURRENT_DATE) - p.start_date) / 7 AS weeks_worked,
       r.hours_per_week * (((COALESCE(p.end_date, CURRENT_DATE) - p.start_date)) / 7) AS total_hours
FROM project_assignment AS pa
JOIN role AS r
    ON pa.role_name = r.role_name
JOIN project AS p
    ON pa.project_id = p.project_id
WHERE pa.staff_id = 'A001' AND p.project_id = 'P0001';
```

project_id	title	staff_id	hours_per_week	start_date	end_date	weeks_worked	total_hours
P0001	AI-Based Smart Campus	A001		2025-03-10	2026-02-01	46	920

Query Specifications:

Who will use the query:

Project Coordinators and R&D Directorate Managers

Purpose of the query:

To determine how much effort a specific academic staff member has contributed to a particular project.

What the query displays:

This query shows the project details, weekly working hours, project duration in weeks, and the total calculated hours worked by the selected staff member for the specified project.

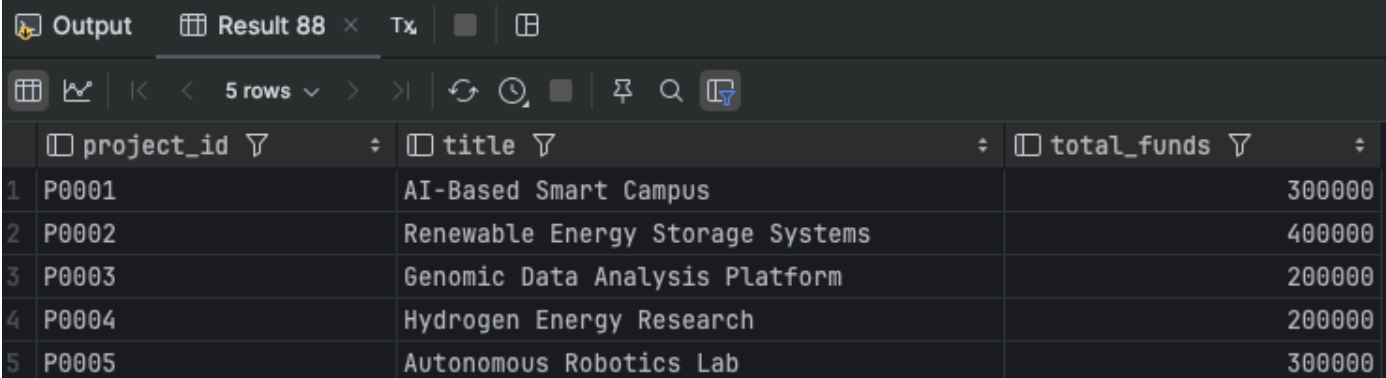
QUERY 6: TOTAL FUNDS RECEIVED PER PROJECT

Query:

```
SELECT p.project_id, p.title, SUM(f.amount) AS total_funds
FROM project AS p
LEFT JOIN fund AS f
    ON p.project_id = f.project_id
GROUP BY p.project_id, p.title
ORDER BY p.project_id;
```

Output:

```
--TOTAL FUNDS RECEIVED PER PROJECT
SELECT p.project_id, p.title, SUM(f.amount) AS total_funds
FROM project AS p
LEFT JOIN fund AS f
    ON p.project_id = f.project_id
GROUP BY p.project_id, p.title
ORDER BY p.project_id;
```



The screenshot shows a database query result interface. At the top, there's a toolbar with icons for 'Output', 'Result 88', 'Tx', and a grid icon. Below the toolbar, there's a navigation bar with icons for table, chart, and a dropdown menu showing '5 rows'. To the right of the dropdown are icons for refresh, zoom, and other controls. The main area displays a table with 3 columns: 'project_id', 'title', and 'total_funds'. The table has 5 rows of data.

	project_id	title	total_funds
1	P0001	AI-Based Smart Campus	300000
2	P0002	Renewable Energy Storage Systems	400000
3	P0003	Genomic Data Analysis Platform	200000
4	P0004	Hydrogen Energy Research	200000
5	P0005	Autonomous Robotics Lab	300000

Query Specifications:

Who will use the query:

R&D Directorate Managers and Financial Officers

Purpose of the query:

To monitor how much funding each project has received.

What the query displays:

This query displays each project along with the total amount of funds received, aggregated from all associated funding records.

QUERY 7: TOTAL EXPENSES PER PROJECT WITH COMPARISON TO TOTAL BUDGET

Query:

```
SELECT p.project_id, p.title, p.status, p.total_budget, SUM(e.amount) AS
total_expenses,
    ROUND(SUM(e.amount)/p.total_budget * 100, 0) AS perct_budget_spent,
    (p.total_budget - SUM(e.amount)) AS remaining_budget
FROM expense AS e
LEFT JOIN project AS p
    ON e.project_id = p.project_id
GROUP BY p.project_id, p.title, p.total_budget
ORDER BY p.project_id;
```


Output:

```
--TOTAL EXPENSES PER PROJECT WITH COMPARISON TO TOTAL BUDGET
SELECT p.project_id, p.title, p.status, p.total_budget, SUM(e.amount) AS total_expenses,
       ROUND(SUM(e.amount)/p.total_budget * 100, 0) AS perct_budget_spent,
       (p.total_budget - SUM(e.amount)) AS remaining_budget
FROM expense AS e
LEFT JOIN project AS p
  ON e.project_id = p.project_id
GROUP BY p.project_id, p.title, p.total_budget
ORDER BY p.project_id;
```

project_id	title	status	total_budget	total_expenses	perct_budget_spent	remaining_budget
P0001	AI-Based Smart Campus	Active	650000.00	120000	18	530000
P0002	Renewable Energy Storage Systems	Active	500000.00	145000	29	355000
P0003	Genomic Data Analysis Platform	Completed	300000.00	280000	93	20000
P0004	Hydrogen Energy Research	Suspended	900000.00	145000	16	755000
P0005	Autonomous Robotics Lab	Active	400000.00	210000	53	190000

Query Specifications:

Who will use the query:
R&D Directorate Managers and Financial Officers

Purpose of the query:
To monitor project expenditures and compare them against approved budgets.

What the query displays:
This query shows the project details, total budget, total expenses, percentage of the budget spent, and remaining budget for each project.

QUERY 8: ALL DELIVERABLES WHICH WERE APPROVED

Query:

```
SELECT *
FROM deliverable
WHERE approval_status = 'Approved';
```

Output:

```
--ALL DELIVERABLES WHICH WERE APPROVED
SELECT *
FROM deliverable
WHERE approval_status = 'Approved';
```

deliverable_id	title	type	submission_date	approval_status	project_id
D00001	AI System Architecture Report	Progress Report	2024-06-01	Approved	P0001
D00004	Energy Storage Performance Study	Final Report	2024-01-20	Approved	P0002
D00005	Battery Control Algorithm	Software	2024-03-10	Approved	P0002
D00006	Genomic Data Analysis Paper	Journal Article	2023-11-05	Approved	P0003
D00007	Genomic Analysis Software Toolkit	Software	2023-05-30	Approved	P0003
D00009	Hydrogen Storage Materials Review	Progress Report	2023-06-25	Approved	P0004
D00010	Robotics Lab Setup Report	Progress Report	2024-07-15	Approved	P0005

Query Specifications:

Who will use the query:
R&D Directorate Managers and Reporting Officers

Purpose of the query:

To identify all deliverables that have been approved and are ready for reporting to funding bodies.

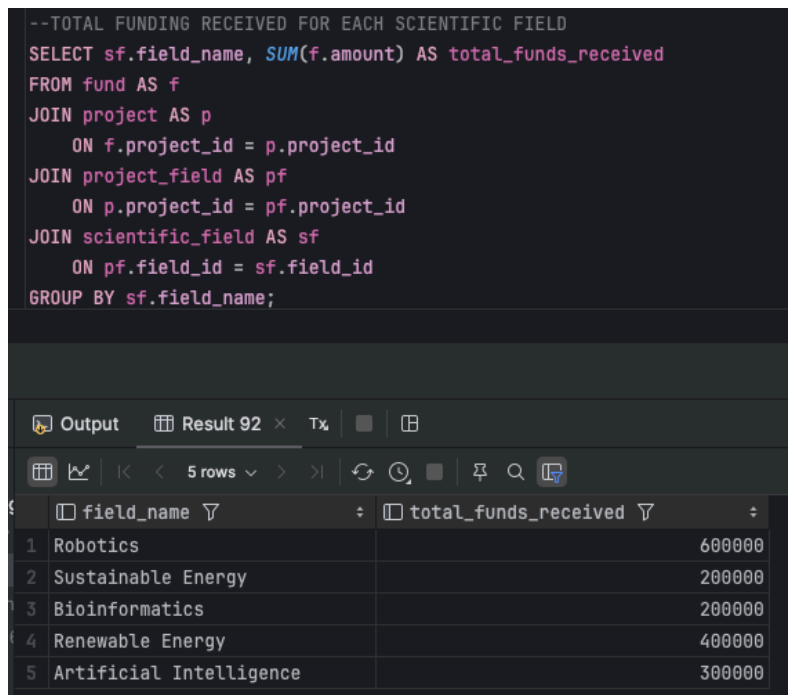
What the query displays:

The query displays all deliverables whose approval status is marked as Approved.

QUERY 9: TOTAL FUNDING RECEIVED FOR EACH SCIENTIFIC FIELD

Query:

```
SELECT sf.field_name, SUM(f.amount) AS total_funds_received
FROM fund AS f
JOIN project AS p
  ON f.project_id = p.project_id
JOIN project_field AS pf
  ON p.project_id = pf.project_id
JOIN scientific_field AS sf
  ON pf.field_id = sf.field_id
GROUP BY sf.field_name;
```

Output:

The screenshot shows a database query execution interface. At the top, the SQL query is displayed in a dark-themed editor. Below the query, the output is shown in a table with 5 rows. The table has two columns: 'field_name' and 'total_funds_received'. The data rows are: Robotics (600000), Sustainable Energy (200000), Bioinformatics (200000), Renewable Energy (400000), and Artificial Intelligence (300000). The interface includes a toolbar with icons for output, result sets, and navigation.

```
--TOTAL FUNDING RECEIVED FOR EACH SCIENTIFIC FIELD
SELECT sf.field_name, SUM(f.amount) AS total_funds_received
FROM fund AS f
JOIN project AS p
  ON f.project_id = p.project_id
JOIN project_field AS pf
  ON p.project_id = pf.project_id
JOIN scientific_field AS sf
  ON pf.field_id = sf.field_id
GROUP BY sf.field_name;
```

field_name	total_funds_received
Robotics	600000
Sustainable Energy	200000
Bioinformatics	200000
Renewable Energy	400000
Artificial Intelligence	300000

Query Specifications:**Who will use the query:**

R&D Directorate Managers and University Management

Purpose of the query:

To analyze research funding distribution across different scientific fields.

What the query displays:

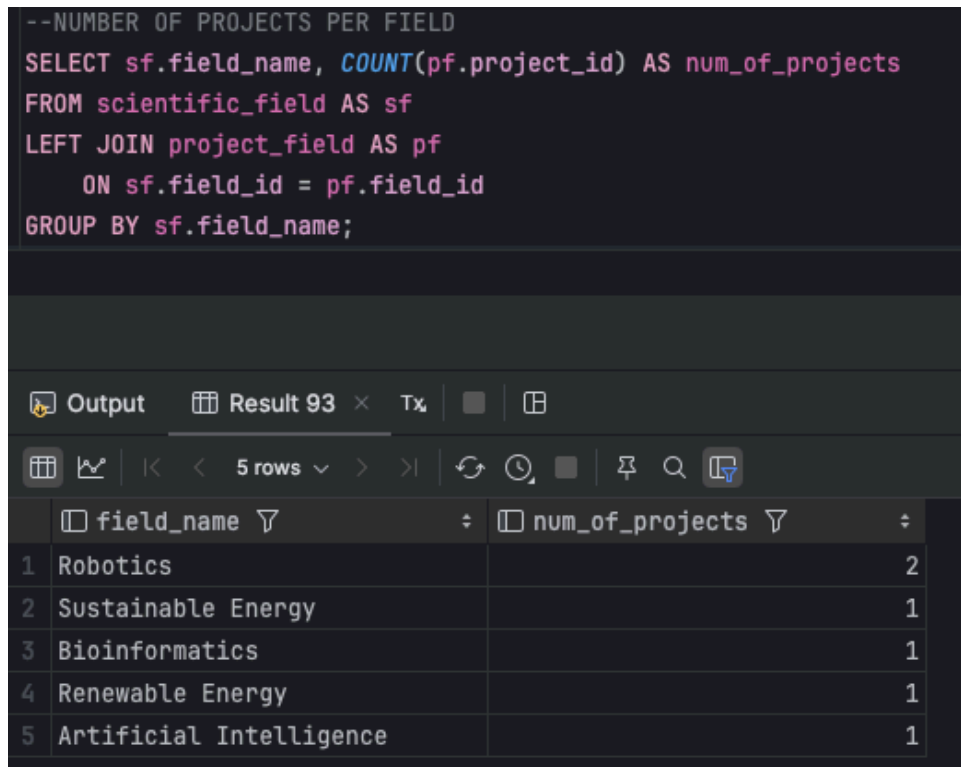
This query shows each scientific field along with the total funding amount received by projects associated with that field.

QUERY 10: NUMBER OF PROJECTS PER FIELD

Query:

```
SELECT sf.field_name, COUNT(pf.project_id) AS num_of_projects
FROM scientific_field AS sf
LEFT JOIN project_field AS pf
    ON sf.field_id = pf.field_id
GROUP BY sf.field_name;
```

Output:



The screenshot shows a database interface with a query editor and a results table. The query is:
--NUMBER OF PROJECTS PER FIELD
SELECT sf.field_name, COUNT(pf.project_id) AS num_of_projects
FROM scientific_field AS sf
LEFT JOIN project_field AS pf
 ON sf.field_id = pf.field_id
GROUP BY sf.field_name;
The results table has two columns: field_name and num_of_projects. It contains five rows of data.

	field_name	num_of_projects
1	Robotics	2
2	Sustainable Energy	1
3	Bioinformatics	1
4	Renewable Energy	1
5	Artificial Intelligence	1

Query Specifications:

Who will use the query:

University Management and R&D Directorate.

Purpose of the query:

To evaluate research activity across different scientific disciplines.

What the query displays:

The query displays each scientific field name and the number of projects associated with that field.

QUERY 11: PROJECTS WHERE THERE ARE MORE THAN ONE FUNDING BODY

Query:

```
WITH num_fb_per_project AS (
    SELECT f.project_id, COUNT(DISTINCT f.funding_body_id) AS num_funding_body
    FROM fund AS f
    GROUP BY f.project_id
)
SELECT p.project_id, p.coordinator_id, p.title, p.status, cte.num_funding_body
```

```
FROM project AS p
JOIN num_fb_per_project AS cte
  ON p.project_id = cte.project_id
WHERE cte.num_funding_body > 1;
```

Output:

```
--PROJECTS WHERE THERE ARE MORE THAN ONE FUNDING BODY
WITH num_fb_per_project AS(
  SELECT f.project_id, COUNT(DISTINCT f.funding_body_id) AS num_funding_body
  FROM fund AS f
  GROUP BY f.project_id
)
SELECT p.project_id, p.coordinator_id, p.title, p.status, cte.num_funding_body
FROM project AS p
JOIN num_fb_per_project AS cte
  ON p.project_id = cte.project_id
WHERE cte.num_funding_body > 1;
```

Output				
Result 94 x Tx				
1 row				
project_id	coordinator_id	title	status	num_funding_body
P0001	A001	AI-Based Smart Campus	Active	2

Query Specifications:

Who will use the query:

R&D Directorate Managers

Purpose of the query:

To identify projects that receive funding from multiple funding bodies.

What the query displays:

This query lists projects along with their coordinator, status, and the number of distinct funding bodies funding each project, showing only those projects funded by more than one funding body.

AI Usage Statement:

Generative AI tools were used to generate synthetic sample data for populating database tables during testing. All database design, schema creation, and SQL queries were developed independently by the team.